



Sustainable Travel in Europe: Making trains more competitive

Capstone project presentation

Practical Data Science

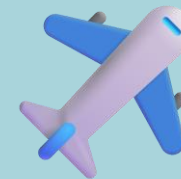
20.05.2026

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Idea and motivation

Relatable:

When you want to take the train...
...but it takes so much longer
...and is more expensive than a flight.



Idea and motivation

Transportation accounts for approximately

16%

of global annual GHG emissions

Source: Our World in Data 2022

Air Travel in Europe accounts for approximately

4%

of EU annual GHG emissions

Source: European Commission 2022

Idea and motivation

What if we consider...

- actual travel time? → Time at airport and station
- actual cost? → Baggage fees (budget airlines)
- the damage of CO₂ emissions? → Social Cost of Carbon

→ Generalised cost

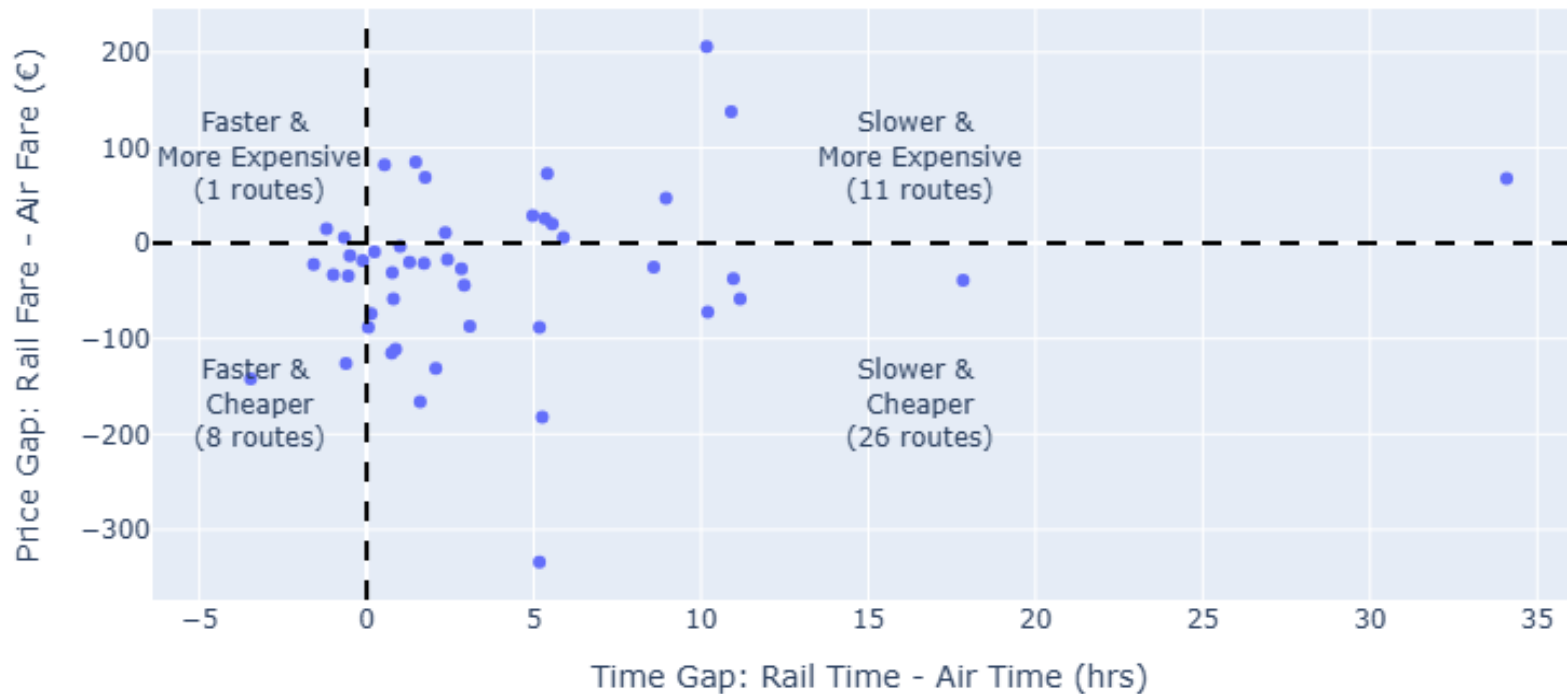
Approach

- Manual data collection from Rome2Rio (46 routes)
- Comparability: same date and decision rule (App. 3)
- Limitations
 - Data availability (API availability)
 - Route choice
 - Decision model

INSIGHTS

Besides being slower, rail is also more expensive on 11 routes

Scenario 3



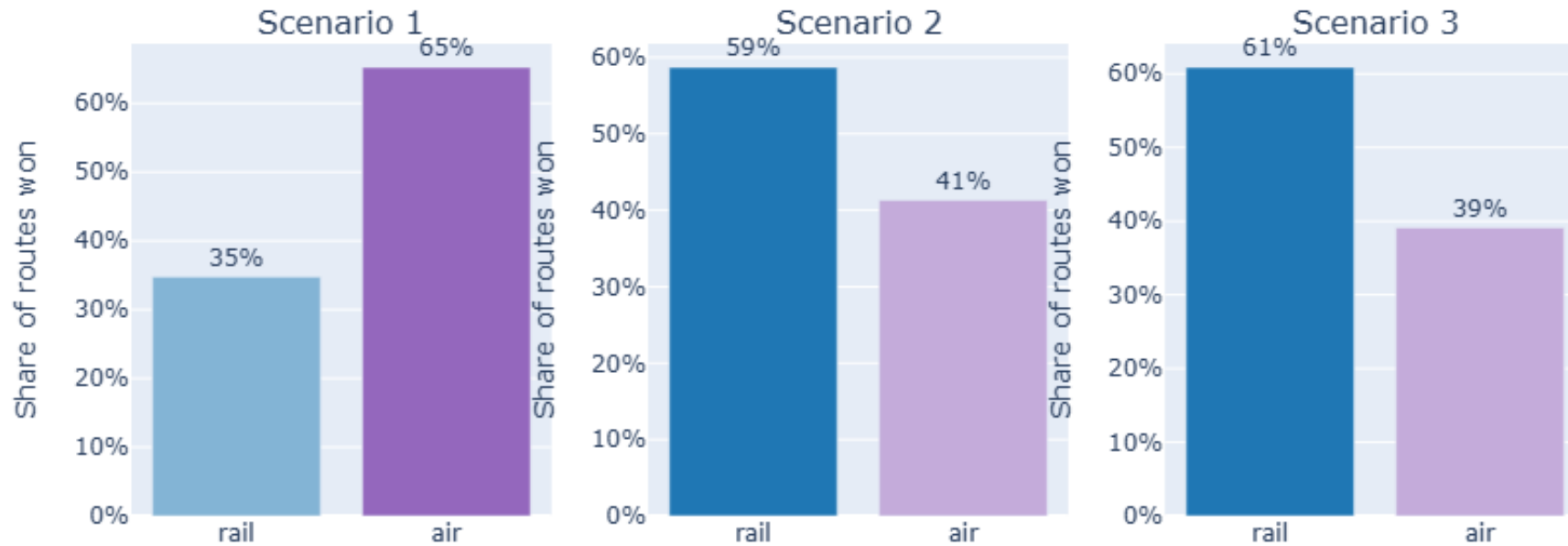
Scenario 1: How you would plan your trip
No airport or station overhead, no baggage fee, no carbon price

Scenario 2: More realistic assumptions
Airport overhead 3h, station overhead 0.5h, baggage fee = 23€, no carbon

Scenario 3
Scenario 2 with inclusion of carbon (150€ per ton)

INSIGHTS

Depending on the scenario, the overall winner flips from air to rail



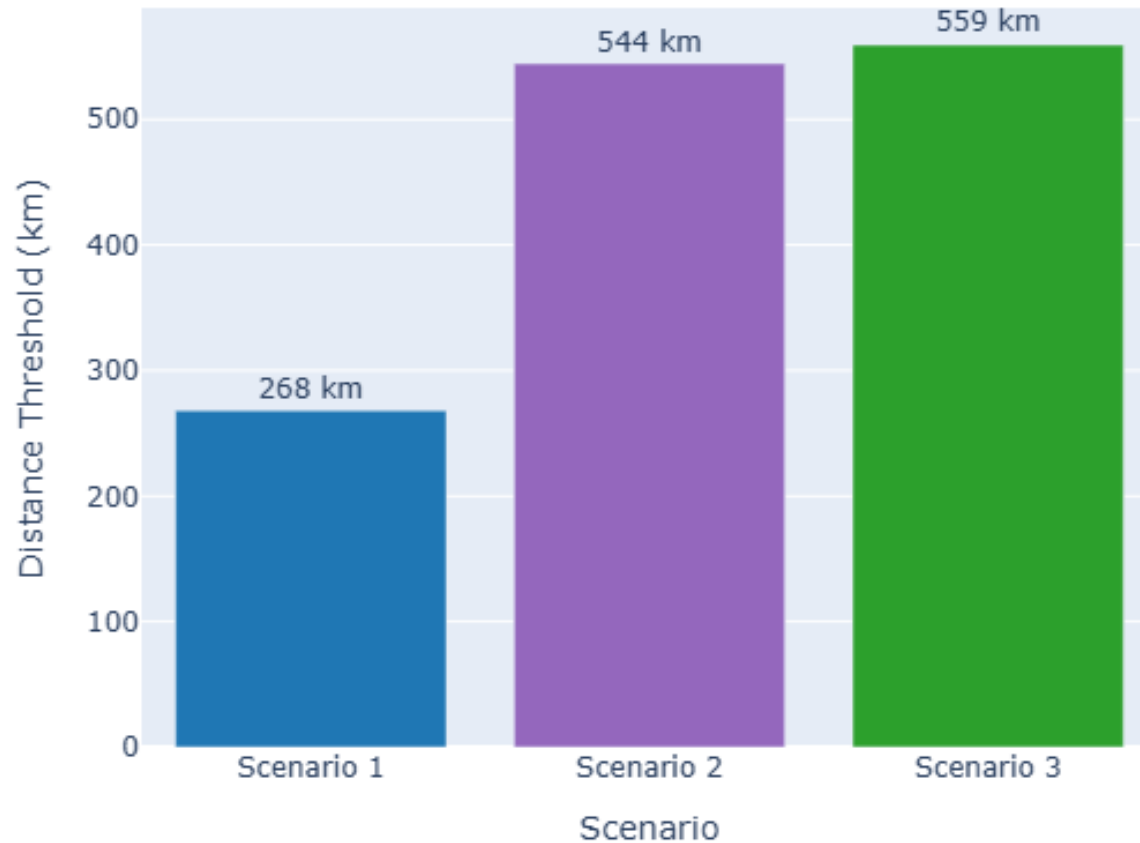
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INSIGHTS

Rail Could Be No. 1 Means of Transport On Middle-distance Routes
Distance Threshold Below Which Rail Is On Average More Cost-effective Than Air



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INSIGHTS

Start with these low-hanging fruits (air <50€ cheaper than train)

Scenario 3 assumptions

Belgrade-Warsow	Berlin-Copenhagen,
Budapest-Bucharest,	Munich-Rome,
Paris-Nice,	Rome-Catania,
Sofia-Istanbul,	Stockholm-Copenhagen

When travelling each of these routes once...

Number of target routes

8

CO2 emitted by air

815.7 kg

CO2 emitted by rail

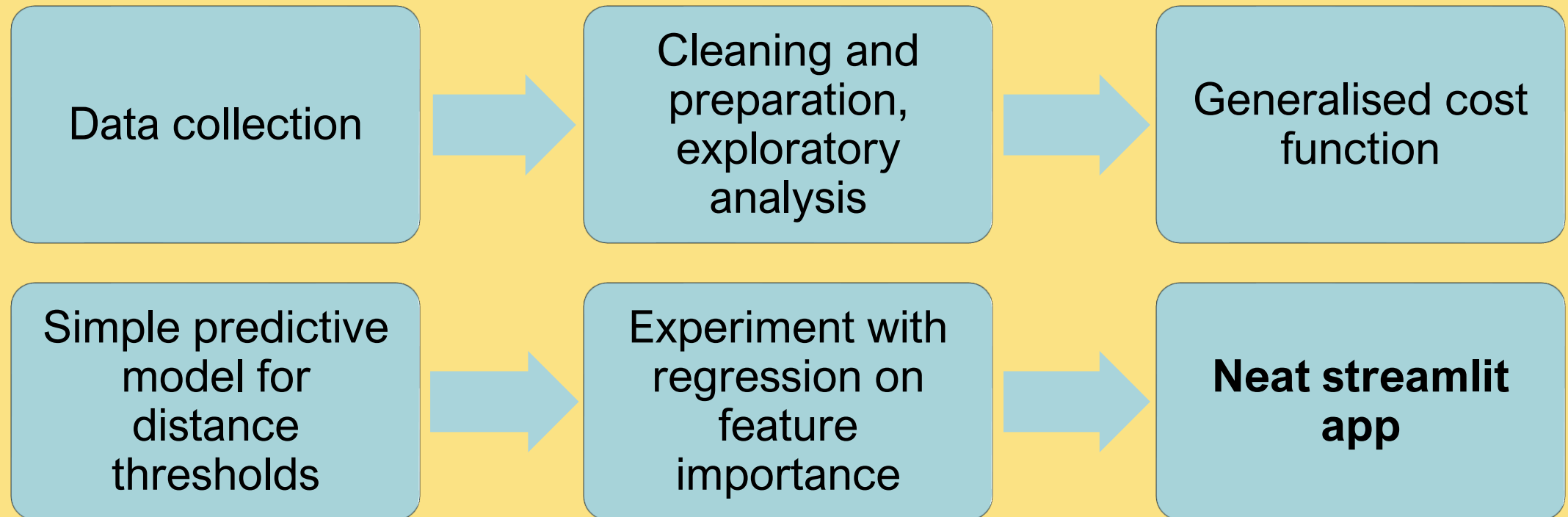
27.7 kg

Share of CO2 saved

96.6%

by switching these 8 routes
to rail

Workflow



Further analysis results

- Logistic regression: time gap overall more important than price gap
- Decision tree: for short-medium distances (up to 520km), price gap matters before time gap

Use cases

- **For politicians:** to guide their policies for green transformation, e.g. effect of price signals
- **For transport planners:** which rail routes to target first
- **For the public:** to make better informed travel decisions

Call to Action

- Save 96.6% of emissions by improving rail offer on 8 target routes!
- Further electrify the grid to save 99% of emissions of 8 target routes!
- For short-medium routes, ensure competitive train prices
- Overall, improve rail speed

Limitations & next steps

- Simplifications
- Data
- Advanced analysis



**Thank you for your attention
and suggestions!**

Appendix

Features of the app

Interactive sliders to adjust assumptions

Predictive model about the distance threshold when rail becomes competitive over air

Calculator of carbon emissions saving by replacing flights by train travel

URL: https://capstone-project--sustainable-travel-ht4rtfeelc62qjfrypq2jo.streamlit.app/Visuals_and_Insights

1. Logistic Regression Model on „winner“

What factor helps us best predict if we choose rail or air on a route?

Remarks on the model:

- „Winner“ as binary dependent variable (rail = 1)
- Gaps defined as rail – air
(gap more positive → rail more expensive/takes more time)
- Only distance predicts already quite well
- Full model (+ price and time gap) has 100% because it doesn't have d.o.f.

Scenario 1 (no overheads, no baggage fee, no carbon)

```
Distance only: 74%  
{'distance_km': np.float64(-0.005927505034449552)}  
  
Full model: 100%  
{'distance_km': np.float64(-0.0045565692349472334),  
  'price_gap': np.float64(-0.18170053525938162),  
  'time_gap': np.float64(-1.341834563253465)}
```

Scenario 3 (air overhead 3h, station overhead 0.5h, baggage fee 23€, carbon 150€/ton)

```
Distance only: 85%  
{'distance_km': np.float64(-0.010143595470516046)}  
  
Full model: 100%  
{'distance_km': np.float64(-0.02606509482330821),  
  'price_gap_adjusted': np.float64(-0.13700813678524074),  
  'time_gap_adjusted': np.float64(-1.0358958134011982)}
```

→ Time gap is more important than price gap in both scenarios.

1. Which means of transport will „win“ for a route?

Logistic Regression Model

Scenario 3 with standardised units (when time gap increases by one standard deviation...

```
time_gap_adjusted    -1.832680
price_gap_adjusted   -1.686186
distance_km          -1.354215
dtype: float64
```

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2. Decision Tree

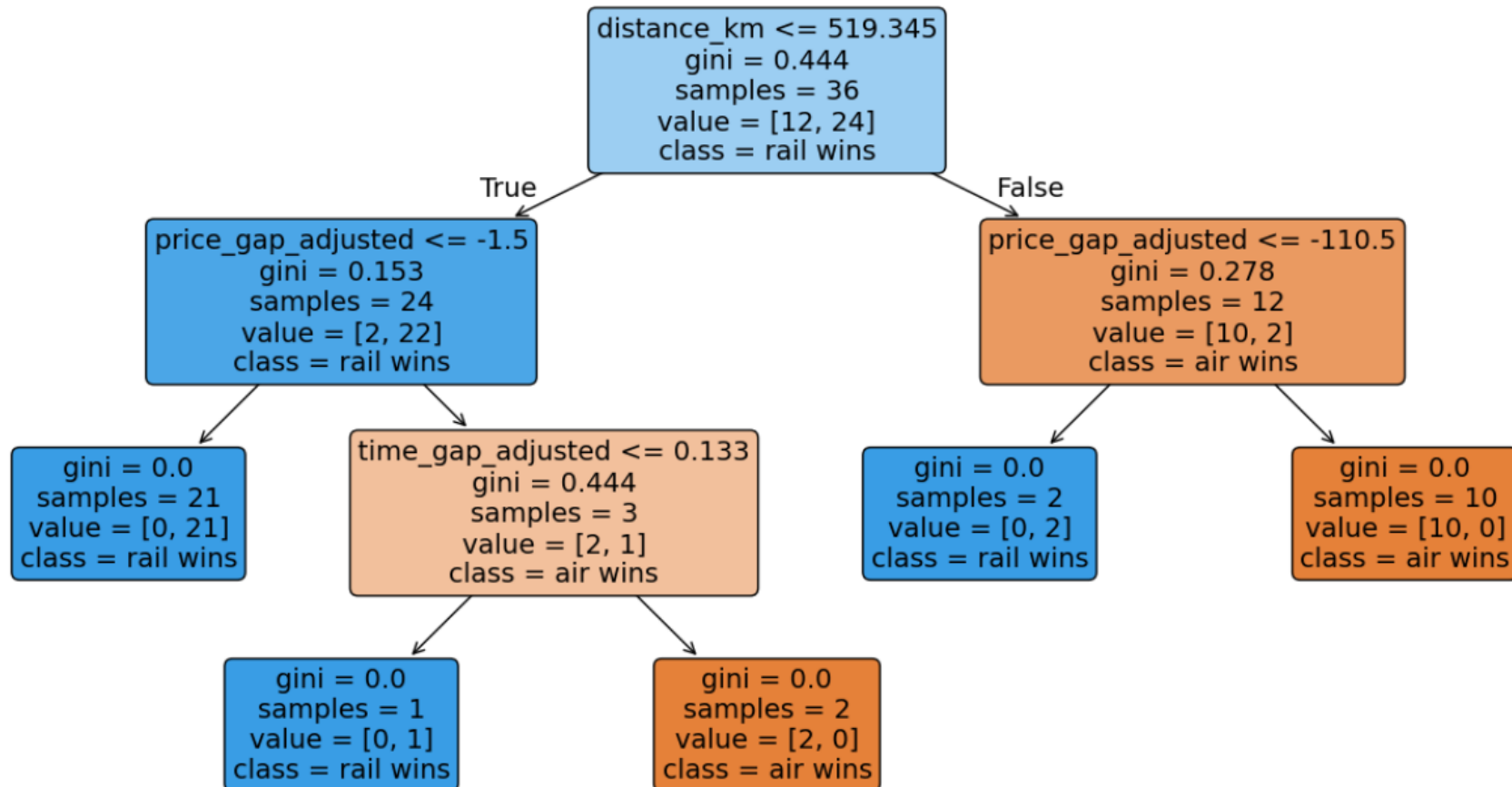
Accuracy: 0.9

distance_km 0.562500

price_gap_adjusted 0.354167

time_gap_adjusted 0.083333

dtype: float64



3. Decision Rule For Data Collection

- All fares for 14 July 2026 (as if booking a trip 3 months in advance)
- Cheapest direct connection. Consider only high-speed rail.
- All travel times must be between 6am and 10pm
- If only price span was available, look price up on train operator's website, else use the average of the price span.